Task 3 – Create Your First Local Repository

Git & GitHub Fundamentals (Industry Workflow) | Estimated Time: 1 Hour

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Step 1 – Create the Project Folder

Create a new directory called `DevOps-Git` and navigate into it:

```
mkdir DevOps-Git
cd DevOps-Git
```

mkdir *creates the folder.* ***cd*** *changes your working directory into it — all subsequent commands run inside this folder.*

Step 2 – Initialize the Repository

Turn the folder into a Git repository:

```
git init
```

This creates a hidden `.git/` subdirectory that stores all version-control data (commits, branches, history). Without this step, Git commands will not work in the folder. You should see: Initialized empty Git repository in .../DevOps-Git/.git/

Step 3 – Create the Required Files

Create three files inside the repository. The quickest way on Ubuntu is the `touch` command, which creates an empty file:

```
touch README.md
touch notes.txt
touch commands.txt
```

You can also add content to the files right away using `echo`:

```
echo "# DevOps-Git Project" > README.md
echo "My Git learning notes" > notes.txt
echo "Git commands reference" > commands.txt
```

- **README.md** — A Markdown file that describes the project — displayed automatically on GitHub.
- **notes.txt** — A plain-text file for personal notes about what you are learning.
- **commands.txt** — A plain-text file to log the Git commands you use as a quick reference.

Step 4 – Check Repository Status

See the current state of your working directory and staging area:

```
git status
```

You will see all three files listed under Untracked files in red. This means Git has detected them but is not yet tracking their changes. To start tracking them you would run `git add .` *followed by* `git commit -m "Initial commit"` *— covered in the next task.*

What each status indicator means:

- **Untracked files** — Files Git has never seen before — not yet included in any commit.
- **Changes not staged** — Tracked files that have been modified but not yet added to the staging area.
- **Changes to be committed** — Files added with git add — ready to be saved in the next commit.
- **nothing to commit** — Working directory is clean; all changes have been committed.

Quick Command Reference

- `mkdir <name>` — Create a new directory.
- `cd <name>` — Change into a directory.
- `git init` — Initialize a new empty Git repository.
- `touch <file>` — Create a new empty file.
- `ls -la` — List all files including hidden ones (use to confirm .git/ exists).
- `git status` — Show the state of the working directory and staging area.

Deliverable

Take screenshots of the following and submit them:

✓ **Repository initialized — terminal output of** `git init` **confirming the empty repo was created.**

✓ **Files created — terminal showing the three files exist (e.g. output of** `ls -la`**).**

✓ **Git status output — terminal showing** `git status` **listing all three files as untracked.**